

GOSFORD HIGH SCHOOL



Year 12

MATHEMATICS EXTENSION 2

HSC Course Assessment Task #3 June 2011

Time Allowed: 60 minutes + 5 minutes reading time

Instructions:

- There are 3 questions.
- Marks allocated to each question are indicated.
- Answer each question on your own paper showing all necessary working.
- Start each question on a new sheet of paper.
- Calculators may be used.

	Topic	Mark
QUESTION 1:	Integration	/18
QUESTION 2:	Volumes	/12
QUESTION 3:	a, b, d (Conics)	/8
	c (Integration)	/4
TOTAL		/42

QUESTION 1: (18 Marks) Use a separate sheet of paper**Marks**

a. Evaluate

2

$$\int_{-1}^1 x^3 + x^2 \sin x \, dx$$

(Giving a brief reason for your answer)

b. Using the table of standard integrals, show that

2

$$\int_{\frac{1}{4}}^1 \frac{dx}{\sqrt{16x^2 - 1}} = \frac{1}{4} \ln(4 + \sqrt{15})$$

c. Use integration by parts to evaluate

3

$$\int_0^{\pi} e^x \sin x \, dx$$

d. It can be shown that:

2

$$\frac{9-x}{1+x+x^2+x^3} = \frac{4-5x}{1+x^2} + \frac{5}{1+x}$$

(Do not prove this!)

Use this result to find

$$\int \frac{9-x}{1+x+x^2+x^3} \, dx$$

Question 1 continues

Question 1 continued

Marks

- e. By using an appropriate substitution prove that

3

$$\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{12} - \frac{\sqrt{3}}{8}$$

- f. Use the substitution $t = \tan \frac{\theta}{2}$

3

to find

$$\int \frac{1}{1 + \sin \theta} d\theta$$

- g. Find

3

$$\int \frac{3x + 4}{\sqrt{8 - 6x - 9x^2}} dx$$

End of Question 1

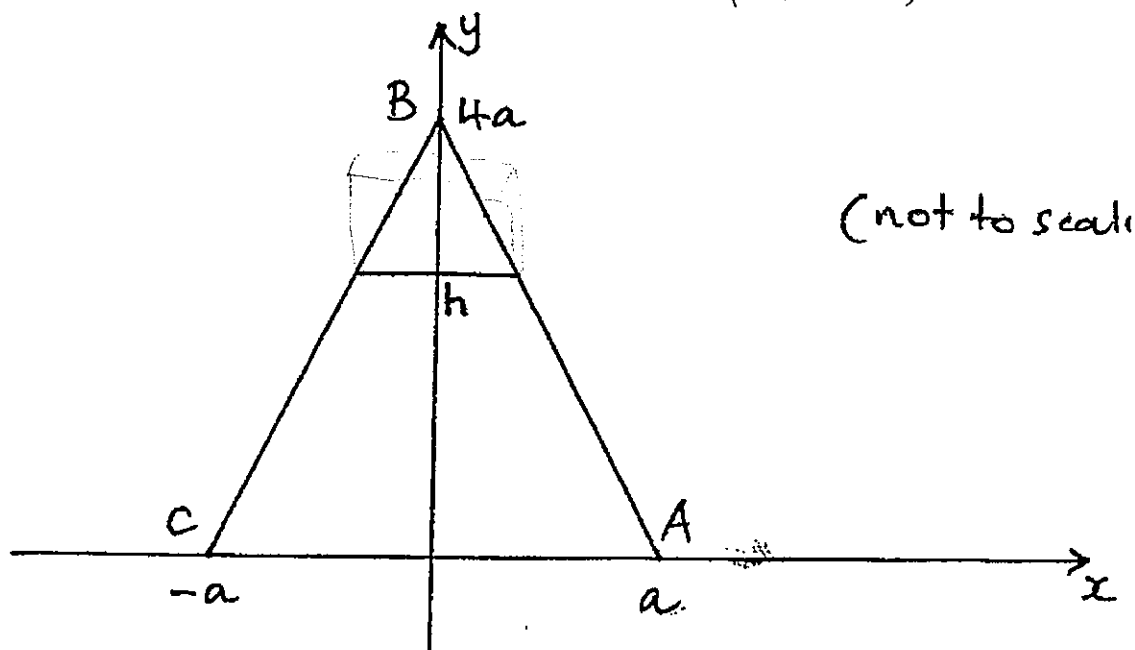
QUESTION 2: (12 Marks) Use a separate sheet of paper

Marks

- a. The region, in the first quadrant, bounded by $y = x(2 - x)^2$, the x axis and $x = 2$ is rotated about the y axis to form a solid. This volume is to be found by the method of cylindrical shells.
- i. Draw a neat diagram to represent the above information. (Indicating a typical cylindrical shell.) 1
- ii. Use this method to find the volume of the solid. 3

- b. The base of a solid is an isosceles triangle ABC as shown:

(Not to scale)



Vertical cross-sections taken at right angles to the y axis are squares. (The line interval indicated at h represents one side of a typical square.)

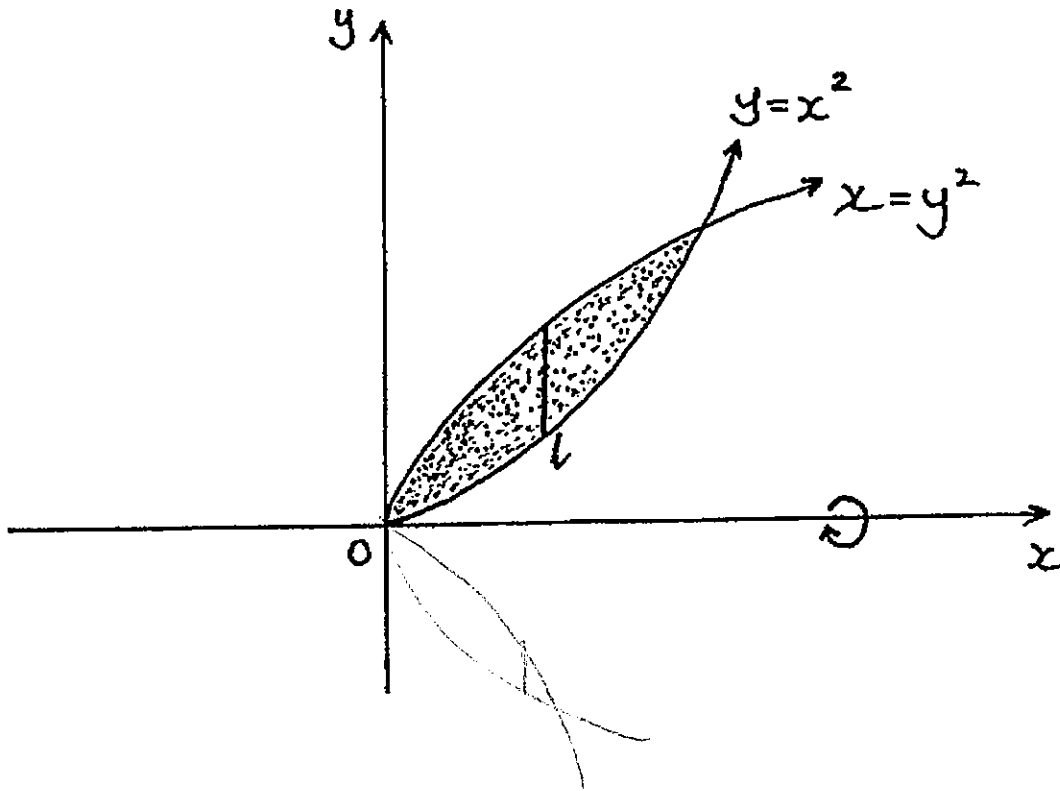
- i. Show that the area of the square cross-section at $y = h$ is given by $4\left(a - \frac{h}{4}\right)^2$ 2
- ii. Hence, find the volume of the solid. 2

Question 2 continues

Question 2 continued

Marks

- c. The shaded region indicated is bounded by the curves $y = x^2$ and $x = y^2$:



This region is rotated about the x axis to form a solid.

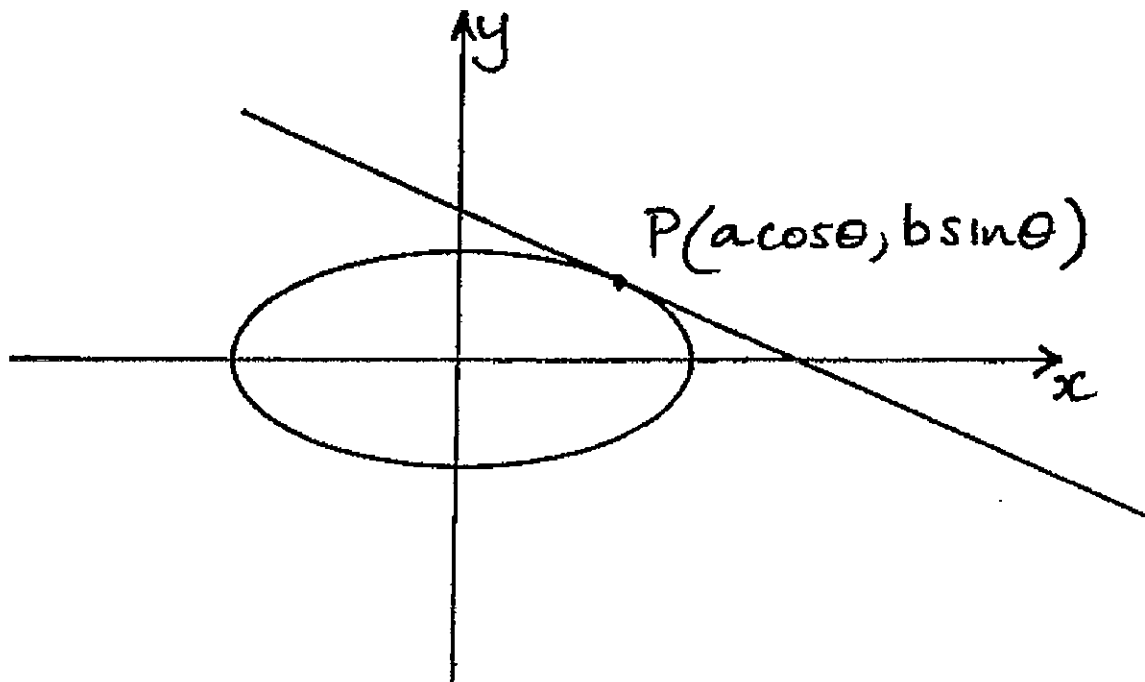
When the region is rotated the vertical line segment l sweeps out an annulus.

- i. By taking annular slices perpendicular to the x axis, express the volume of the resulting solid as an integral. 2
- ii. Evaluate the integral in Part (i) 2

End of Question 2

QUESTION 3: (12 Marks) Use a separate sheet of paper**Marks**a Sketch the ellipse $9x^2 + 4y^2 = 36$, indicating the foci and directrices

2

b Given the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 

If $P(a \cos \theta, b \sin \theta)$ is a variable point on the ellipse show that the tangent through P has equation:

2

$$bx \cos \theta + ay \sin \theta - ab = 0$$

Question 3 continues

Question 3 continued

Marks

c Given that:

$$I_n = \int_1^e (1nx)^n dx$$

where n is a non-negative integer

- i. Prove that $I_n = e - nI_{n-1}$ ($n \neq 0$) 2
- ii. Hence, evaluate I_4 2

- d i. For $y = mx + c$ to be a tangent to the ellipse 2

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \text{ prove that}$$

$$c^2 = a^2m^2 + b^2$$

- ii. Hence, show that the pair of tangents drawn from the point (3,4) to the ellipse 2

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \text{ are at right angles to each other}$$

END OF ASSESSMENT TASK

P1

$$a. \int_{-1}^1 x^3 + x^2 \sin x \, dx = 0$$

since $x^3 + x^2 \sin x$ is an odd function.

$$b. \int_{\frac{1}{4}}^1 \frac{dx}{\sqrt{16x^2 - 1}} = \int_{\frac{1}{4}}^1 \frac{dx}{\sqrt{16(x^2 - \frac{1}{16})}}$$

$$= \frac{1}{4} \int_{\frac{1}{4}}^1 \frac{dx}{\sqrt{x^2 - \frac{1}{16}}}$$

Using table of s.i :

$$= \frac{1}{4} \left[\ln \left(x + \sqrt{x^2 - \frac{1}{16}} \right) \right]_{\frac{1}{4}}^1$$

$$= \frac{1}{4} \left[\ln \left(1 + \sqrt{15} \right) - \ln \left(\frac{1}{4} \right) \right]$$

$$= \frac{1}{4} \left[\ln \left(\frac{4 + \sqrt{15}}{4} \right) + \ln 4 \right]$$

$$= \frac{1}{4} \ln (4 + \sqrt{15})$$

$$\therefore I = \int_0^\pi e^x \sin x \, dx = \int_0^\pi \sin x \cdot e^x \, dx$$

Using integration by parts

$$= \left[\sin x \cdot e^x \right]_0^\pi - \int_0^\pi \cos x \cdot e^x \, dx$$

$$= 0 - \int_0^\pi \cos x \cdot e^x \, dx$$

$$= - \left[\cos x \cdot e^x \right]_0^\pi + \int_0^\pi \sin x \cdot e^x \, dx$$

$$\therefore I = e^\pi + 1 - I$$

$$2I = e^\pi + 1$$

$$I = \frac{e^\pi + 1}{2}$$

$$d. \int \frac{9-x}{1+x+x^2+x^3} \, dx$$

$$= \int \frac{4-5x}{1+x^2} \, dx + \int \frac{5}{1+x} \, dx$$

$$= \int \frac{4}{1+x^2} \, dx - \frac{5}{2} \int \frac{2x}{1+x^2} \, dx + \int \frac{5}{1+x} \, dx$$

$$= 4 \tan^{-1} x - \frac{5}{2} \ln(1+x^2) + 5 \ln|1+x|$$

$$e. \int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} \, dx \quad \left\{ \begin{array}{l} \text{let } x = \sin \theta \\ \frac{dx}{d\theta} = \cos \theta \\ x = \frac{1}{2}, \theta = \frac{\pi}{6} \\ x = 0, \theta = 0 \end{array} \right.$$

$$= \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{\sqrt{1-\sin^2 \theta}} \cdot \cos \theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{\sqrt{\cos^2 \theta}} \cdot \cos \theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \sin^2 \theta \, d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{6}} 1 - \cos 2\theta \, d\theta$$

$$= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) - 0 \right] = \frac{\pi}{12} - \frac{\sqrt{3}}{8}$$

$$f. (i) \quad t = \tan \frac{\theta}{2}$$

$$\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$$

$$\frac{dt}{d\theta} = \frac{1}{2} (1 + \tan^2 \frac{\theta}{2})$$

$$= \frac{1}{2} (1 + t^2)$$

$$(ii) \quad \sin \theta = 2 \sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2}$$

$$= \frac{2 \sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \cdot \cos^2 \frac{\theta}{2}$$

$$= 2 \tan \frac{\theta}{2} \cdot \frac{1}{\sec^2 \frac{\theta}{2}}$$

$$= 2t \cdot \frac{1}{1 + \tan^2 \frac{\theta}{2}}$$

$$= \frac{2t}{1+t^2} \text{ as req.}$$

$$1) \quad \int \frac{1}{1 + \sin \theta} \cdot d\theta \quad t = \tan \frac{\theta}{2}$$

$$= \int \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2} \quad \frac{dt}{d\theta} = \frac{1+t^2}{2}$$

$$\frac{2dt}{1+t^2} = d\theta$$

$$= \int \frac{2dt}{\frac{1+t^2+2t}{1+t^2}}$$

$$= \int \frac{2dt}{(1+t)^2} = 2 \int (1+t)^{-2} dt$$

$$= 2 \left[\frac{(1+t)^{-1}}{-1} \right] + c$$

$$= \frac{-2}{1+t} + c$$

$$= \frac{-2}{1 + \tan \frac{\theta}{2}} + c$$

$$g. \quad \int \frac{3x+4}{\sqrt{8-6x-9x^2}} dx$$

$$= -\frac{1}{6} \int \frac{-18x-6}{\sqrt{8-6x-9x^2}} dx + \int \frac{3}{\sqrt{8-6x-9x^2}} dx$$

$$= I + J$$

$$\therefore I = -\frac{1}{6} \int \frac{-18x-6}{\sqrt{8-6x-9x^2}} dx \quad \text{let } u = 8-6x$$

$$\frac{du}{dx} = -6$$

$$= -\frac{1}{6} \int \frac{du}{\sqrt{u}}$$

$$= -\frac{1}{6} \int u^{-1/2} du$$

$$= -\frac{1}{6} \left[\frac{u^{1/2}}{1/2} \right] + C_1$$

$$= -\frac{1}{3} \sqrt{u} + C_1$$

$$= -\frac{1}{3} \sqrt{8-6x-9x^2} + C_1$$

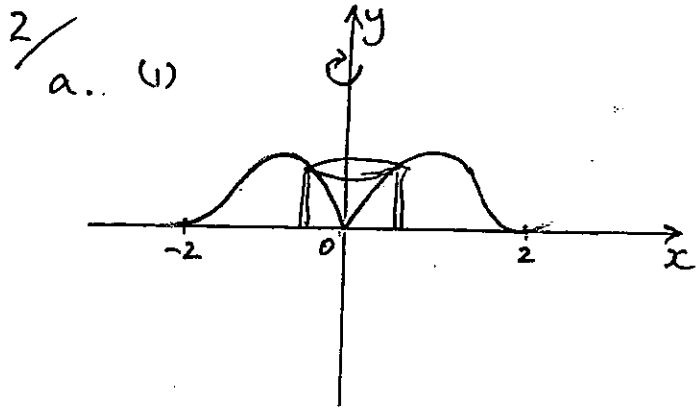
$$J = \int \frac{3}{\sqrt{9\left(\frac{8}{9} - \frac{2x}{3} - x^2\right)}} dx$$

$$= \int \frac{dx}{\sqrt{-(x^2 + \frac{2x}{3} + \frac{1}{9}) + 1}}$$

$$= \int \frac{dx}{\sqrt{1 - \left(x + \frac{1}{3}\right)^2}}$$

$$= \sin^{-1}\left(x + \frac{1}{3}\right) + C_2$$

$$I + J = \sin^{-1}\left(x + \frac{1}{3}\right) - \frac{1}{3}\sqrt{8 - 6x - 9x^2} + C$$



$$(ii) V = 2\pi \int_0^2 x \cdot y \cdot dx$$

$$\text{where } y = x(2-x)^2$$

$$V = 2\pi \int_0^2 x^2 (2-x)^2 dx$$

$$= 2\pi \int_0^2 x^2 (4 - 4x + x^2) dx$$

$$= 2\pi \int_0^2 (4x^2 - 4x^3 + x^4) dx$$

$$= 2\pi \left[\frac{4x^3}{3} - x^4 + \frac{x^5}{5} \right]_0^2$$

$$= 2\pi \left[\left(\frac{32}{3} - 16 + \frac{32}{5} \right) - 0 \right]$$

$$= 2\pi \left(\frac{16}{15} \right) = \frac{32\pi}{15} \text{ units}^3$$

$$b. (i) \text{ Eqn AB} \Rightarrow \frac{x}{a} + \frac{y}{4a} = 1$$

$$4x + y = 4a$$

$$y = -4x + 4a$$

$$\therefore \text{ when } y = h,$$

$$h = -4x + 4a$$

$$4x = 4a - h$$

$$x = a - \frac{h}{4}$$

$$\therefore \text{ Area of square} = \left[2\left(a - \frac{h}{4}\right) \right]^2$$

$$= 4\left(a - \frac{h}{4}\right)^2$$

$$(ii) V = \int_0^{4a} 4\left(a - \frac{h}{4}\right)^2 dh$$

$$= 4 \int_0^{4a} \left(a^2 - \frac{ah}{2} + \frac{h^2}{16} \right) dh$$

$$= 4 \left[a^2 h - \frac{ah^2}{4} + \frac{h^3}{48} \right]_0^{4a}$$

$$= 4 \left[\left(4a^3 - 4a^3 + \frac{4a^3}{3} \right) - 0 \right]$$

$$= \frac{16a^3}{3} \text{ units}^3$$

$$c. (i) A(x) = \pi(R^2 - r^2)$$

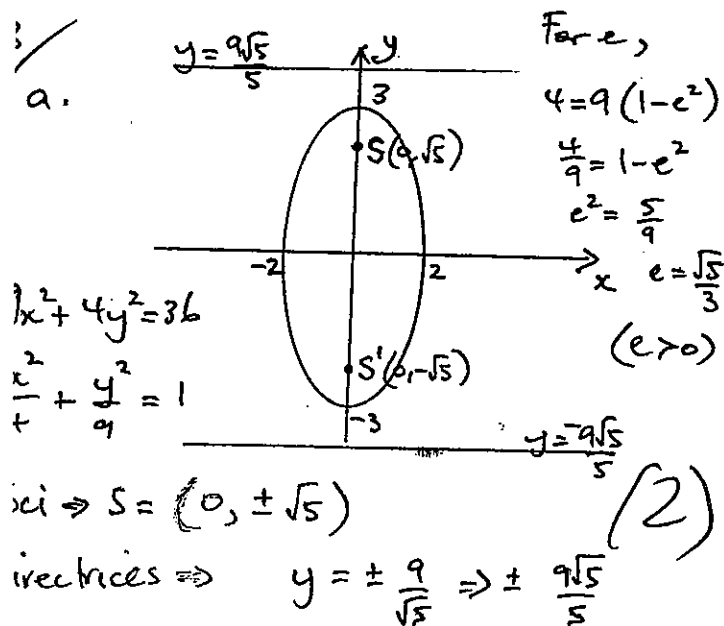
$$= \pi((\sqrt{x})^2 - (x^2)^2)$$

$$= \pi(x - x^4)$$

$$\therefore \Delta V = \pi(x - x^4) \Delta x$$

$$\therefore V = \pi \int_0^1 (x - x^4) dx$$

$$\begin{aligned}
 11) \quad V &= \pi \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 \\
 &= \pi \left[\left(\frac{1}{2} - \frac{1}{5} \right) - 0 \right] \\
 &= \frac{3\pi}{10} \text{ units}^3
 \end{aligned}$$



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{2y}{b^2} \cdot \frac{dy}{dx} = -\frac{2x}{a^2}$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

(at P)

$$\begin{aligned}
 \frac{dy}{dx} &= -\frac{b^2 \cdot a \cos \theta}{a^2 b \sin \theta} \\
 &= -\frac{b \cos \theta}{a \sin \theta}
 \end{aligned}$$

$\therefore$ eqn of tangent at P :

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a y \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$bx \cos \theta + ay \sin \theta - ab(\sin^2 \theta + \cos^2 \theta) = 0$$

$$\therefore bx \cos \theta + ay \sin \theta - ab = 0 \quad (2)$$

as req.

$$c. (1) I_n = \int_1^e (\ln x)^n dx$$

Using integration by parts :

$$\begin{aligned}
 I_n &= \int_1^e (\ln x)^n \cdot 1 dx \\
 &= \left[(\ln x)^n \cdot x \right]_1^e - \int_1^e n \cdot \frac{1}{x} (\ln x)^{n-1} \cdot x dx \\
 &= (e - 0) - n \int_1^e (\ln x)^{n-1} dx
 \end{aligned}$$

$$\therefore I_n = e - n I_{n-1} \quad (2)$$

(11)

$$I_4 = e - 4I_3$$

$$I_3 = e - 3I_2$$

$$I_2 = e - 2I_1$$

Now

$$I_1 = \int_1^e \ln x dx$$

$$\begin{aligned}
 I_1 &= \left[x \ln x \right]_1^e - \int_1^e 1 dx \\
 &= e - (e - 1) \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 u &= \ln x & \frac{du}{dx} &= \frac{1}{x} \\
 v &= x & \frac{dv}{dx} &= 1
 \end{aligned}$$

$$\begin{aligned}
 I_4 &= e - 4I_3 \\
 &= e - 4(e - 3I_2) \\
 &= e - 4e + 12I_2 \\
 &= -3e + 12I_2 \\
 &= -3e + 12(e - 2I_1) \\
 &= -3e + 12e - 24I_1 \\
 &= 9e - 24I_1 \\
 &= 9e - 24(1) \\
 &= 9e - 24
 \end{aligned}$$

Ex. (1) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ — (1)

$y = mx + c$ — (2)

Sub (2) in (1)

$$\frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$$\frac{x^2}{a^2} + \frac{m^2x^2 + 2mcx + c^2}{b^2} = 1$$

$$b^2x^2 + a^2m^2x^2 + 2amcx + a^2c^2 = a^2b^2$$

$$(b^2 + a^2m^2)x^2 + 2amcx + a^2(c^2 - b^2) = 0$$

to be a tangent this equation has one real root

i.e. $\Delta = 0$

$$4a^4m^2c^2 - 4a^2(b^2 + a^2m^2)(c^2 - b^2) = 0$$

$$4a^4m^2c^2 - (4a^2b^2 + 4a^4m^2)(c^2 - b^2) = 0$$

$$4a^4m^2c^2 - 4a^2b^2c^2 + 4a^2b^4 - 4a^4m^2c^2 + 4a^4m^2b^2 = 0$$

$$4a^2b^4 + 4a^4m^2b^2 = 4a^2b^2c^2$$

$$(\div 4a^2b^2) \quad (2)$$

$$\therefore b^2 + a^2m^2 = c^2 \text{ as req}$$

(11) Since (3,4) lies on

$$y = mx + c$$

$$\therefore 4 = 3m + c \Rightarrow 4 - 3m = c$$

and since $c^2 = b^2 + a^2m^2$

$$c^2 = 9 + 16m^2$$

$$\therefore (4 - 3m)^2 = 9 + 16m^2$$

$$16 - 24m + 9m^2 = 9 + 16m^2$$

$$\therefore 7m^2 + 24m - 7 = 0$$

If m_1 and m_2 are the roots of this equation

$$\text{then } m_1m_2 = -\frac{7}{7}$$

$$= -1$$

Hence the two tangents are at right angles 2/

$$\text{Since } m_1m_2 = -1.$$